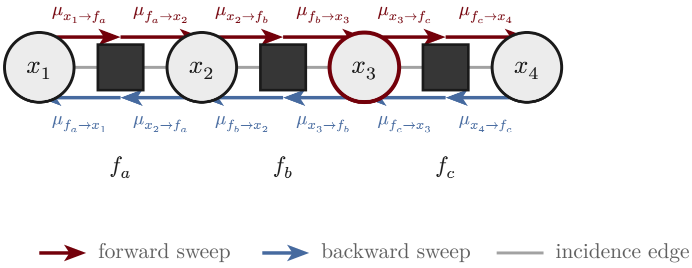
Sum–product message passing

two directed messages per edge of a factor graph; on a chain, a forward sweep ($x_1 \to x_4$) and a backward sweep ($x_4 \to x_1$) give every marginal as the product of its arriving messages



variable $\to$ factor

$\mu_{x\to f}(x) = \prod_{g\in ne(x)\setminus f} \mu_{g\to x}(x)$

factor $\to$ variable

$\mu_{f\to x}(x) = \sum_{x'\setminus x} f(x, x') \prod_{y\in ne(f)\setminus x} \mu_{y\to f}(y)$

marginal (read-out)

$p(x) \propto \prod_{f\in ne(x)} \mu_{f\to x}(x)$